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Enterprise Architect Thinking Guide — Part 2

Question 3: What Can Go Wrong in Migration? How Do You Handle It?

The EA Mindset

"An experienced EA doesn't hope for the best — they plan for the worst. Every migration has risks. The difference between success and failure is whether you identified them BEFORE they became problems."

Migration Risk Registry — 12 Real-World Failure Patterns

Category A: Technical Risks

#	What Goes Wrong	Root Cause	How I Prevent It
1	Application breaks after lift-and-shift	Hidden dependencies on hostname, IP address, local file paths, or hardcoded endpoints	Discovery phase: Run dependency mapping (Application Discovery Service). Document every server-to-server connection. Test in staging environment that mirrors cloud networking.
2	Database migration data loss or corruption	Large databases timeout during migration, CDC lag not monitored, character encoding mismatches	Use AWS DMS with CDC. Full load + continuous replication. Run data validation scripts (row count, checksum, business-rule checks) BEFORE cutover. Always have rollback window.
3	Performance degradation after migration	Wrong instance sizing, different storage IOPS profile (SAN → EBS), network latency between	Right-sizing ≠ same-sizing. Use actual utilization data, not nameplate capacity. Run performance benchmarks in

#	What Goes Wrong	Root Cause	How I Prevent It
		tiers that were previously co-located	cloud BEFORE cutover. Use provisioned IOPS EBS where needed.
4	DNS/Network connectivity failures at cutover	DNS TTL not lowered in advance, firewall rules incomplete, VPN/Direct Connect not tested under load	Lower DNS TTL to 60 seconds 48 hours before cutover. Test all network paths in advance. Have a network engineer on the bridge during cutover.
5	License compliance violations	Oracle/SQL Server licensing doesn't transfer to cloud the same way, bringing BYOL to wrong instance types	License audit during discovery phase. Engage vendor licensing specialist. Understand BYOL rules per cloud provider. Consider license-included or open-source alternatives (Oracle → Aurora PostgreSQL).

Category B: Organizational Risks

#	What Goes Wrong	Root Cause	How I Prevent It
6	Team skill gap causes delays	Engineers unfamiliar with cloud networking, IAM, or IaC — every task takes 3x longer	Invest in training BEFORE migration starts. Embed cloud-experienced engineers (AWS ProServ or partner) for Wave 1. Knowledge transfer is explicit deliverable.
7	Application owners unavailable for testing	Business teams don't prioritize UAT, or key person is on leave during cutover window	Book UAT windows 6 weeks in advance. Get named testers with backup. UAT is a GATE — no sign-off = no cutover. Include in project RACI.

#	What Goes Wrong	Root Cause	How I Prevent It
8	Scope creep — "while we're migrating, let's also..."	Stakeholders add modernization, refactoring, or feature requests mid-migration	Hard boundary: Migration is migration. Modernization is Phase 2. Mixing them doubles timeline and risk. Document this in the SOW.

Category C: Cost Risks

#	What Goes Wrong	Root Cause	How I Prevent It
9	Cloud costs 2-3x higher than estimated	Lifted same-sized servers without right-sizing, forgot data transfer costs, NAT Gateway charges, unused resources after migration	Right-size based on utilization data (not spec). Budget for data transfer (egress during migration period). Set up AWS Budgets and Cost Anomaly Detection from Day 1. Monthly FinOps review.
10	Dual-run costs exceed budget	Running both on-prem and cloud simultaneously during migration takes longer than planned	Plan for 3-month dual-run buffer in budget. Decommission migrated on-prem servers within 30 days of successful cutover. Track dual-run cost weekly.

Category D: Security & Compliance Risks

#	What Goes Wrong	Root Cause	How I Prevent It
11	Security posture weaker in cloud than on-prem	Security team not involved early, default VPC used, overprivileged IAM roles, no encryption	Security architect involved from Day 1. Landing zone includes: private subnets, no public access, KMS encryption mandatory, least-privilege IAM, GuardDuty + Config enabled. Security is pre-requisite, not afterthought.

#	What Goes Wrong	Root Cause	How I Prevent It
12	Regulatory non-compliance post-migration	Data moved to non-approved region, audit logs not configured, retention policies not applied	Compliance checklist as migration gate. Verify data residency, encryption, audit logging, retention policies BEFORE each wave cutover. Security Hub compliance dashboard from Day 1.

How I Handle Issues When They Occur

Triage Framework:

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Issue Detected
├── Severity? (Critical / High / Medium / Low)
├── Can we fix it within the cutover window?
│   ├── YES → Fix, retest, proceed
│   └── NO → Rollback to on-prem, schedule retry
├── Is it a pattern that will affect future waves?
│   ├── YES → Root cause analysis, update migration playbook
│   └── NO → Isolated issue, document and move on
└── Communication
    ├── SEV1/2 → Immediate stakeholder notification
    └── SEV3/4 → Include in weekly status report

```

Real Example from My Experience:

At Advantest, during the GLP application migration to AWS, we discovered that the application had hardcoded internal DNS names that didn't resolve in the cloud VPC. We caught this during staging testing (not production cutover) because we had mandated a full regression cycle in the cloud environment before cutover. Fix: Route 53 private hosted zone with the same DNS names pointing to cloud resources. Lesson: Always test name resolution end-to-end.

Question 4: How Do You Ensure Timelines When You Have External Dependencies?

The EA Mindset

"As an EA, I don't just manage MY team's timeline — I manage the critical path across ALL teams. Dependencies are the #1 cause of enterprise project delays. My job is to identify them early, track them relentlessly, and escalate before they become blockers."

Dependency Management Framework

Step 1: Dependency Mapping (During Planning Phase)

Dependency	Owner	Lead Time	Risk Level	Mitigation
VPN/Direct Connect provisioning	Network Team	6-8 weeks	HIGH	Order in Week 1 of project, not Week 6
Firewall rule changes	Network Security	2-3 weeks per request	MEDIUM	Batch all rules for a wave into one change request
SSL certificate provisioning	Security Team	1-2 weeks	LOW	Use ACM (free, auto-renewing) instead of manual certs
DNS zone delegation	Network Team	1-2 weeks	MEDIUM	Request early, test in staging first
Compliance sign-off	GRC Team	4-6 weeks	HIGH	Engage in Week 1, provide documentation template upfront

Dependency	Owner	Lead Time	Risk Level	Mitigation
AWS SES production access	AWS	2-4 weeks	MEDIUM	Apply early, use SNS as fallback for critical alerts
AWS account limits increase	AWS Support	1-3 days (with Enterprise Support)	LOW	Request before Wave 1, not during
Penetration testing approval	Security + AWS	2-3 weeks	MEDIUM	Schedule 4 weeks before go-live

Step 2: Critical Path Management

Week 1-2: Foundation Phase

- └─ [CRITICAL PATH] Order Direct Connect / VPN ← 6-8 week lead time
- └─ [CRITICAL PATH] Submit compliance documentation to GRC ← 4-6 weeks
- └─ Request AWS SES production access ← 2-4 weeks
- └─ Request account service limit increases
- └─ Set up landing zone (parallel – no dependencies)

Week 3-6: Build Phase

- └─ Terraform modules, CI/CD pipelines (no dependencies)
- └─ [TRACK] VPN/DC provisioning status – weekly check-in with network team
- └─ [TRACK] GRC compliance review progress
- └─ Security architecture review (engage security team)

Week 7-8: Integration Phase

- └─ [GATE] VPN/Direct Connect must be live ← If not, escalate to CTO
- └─ End-to-end connectivity testing
- └─ Firewall rules validation

Week 9+: Migration Waves

- └─ [GATE] Compliance sign-off must be received ← If not, delay wave
- └─ Execute with tested, validated dependencies

Step 3: RAID Log (Risks, Assumptions, Issues, Dependencies)

I maintain a **live RAID log** reviewed in every weekly steering committee:

ID	Type	Description	Owner	Due Date	Status	Escalation
D-001	Dependency	Direct Connect provisioning	Network Lead	Week 6	 In Progress	Escalate if not ordered by Week 2
D-002	Dependency	GRC compliance sign-off	Compliance Manager	Week 8	 At Risk	Meeting scheduled with GRC VP
R-003	Risk	AWS SES production approval delay	Cloud Team	Week 4	 On Track	Fallback: use SNS for alerts
D-004	Dependency	Firewall rules for Wave 1	Network Security	Week 7	 Submitted	Follow up if no response by Week 5

Step 4: Specific Dependency Handling Strategies

Networking Team Delays:

- **Prevention:** Include network team in kickoff meeting, present FULL list of requirements upfront (all VPN tunnels, all firewall rules, all DNS changes for ALL waves)
- **Escalation path:** If network work is > 1 week late → escalate to Infrastructure VP with impact statement: "Migration Wave 1 will slip by 2 weeks, affecting \$X in delayed savings"
- **Workaround:** For development/testing, use AWS Site-to-Site VPN (can be provisioned in hours) while waiting for Direct Connect

Compliance/Regulation Team Delays:

- **Prevention:** Provide a pre-filled compliance template with all answers ready — make their job easy
- **Parallel processing:** Request conditional approval for non-production waves while full approval is pending

- **Escalation:** Frame as business risk — "Without compliance sign-off by date X, we miss the data center lease deadline, committing to \$2.4M/year"

AWS Service Delays (SES, limit increases, etc.):

- **Prevention:** Submit all service requests in Week 1 — don't wait until you need them
 - **Enterprise Support:** Use the TAM (Technical Account Manager) to expedite requests
 - **Workaround:** Always have a Plan B:
 - SES delayed → Use SNS for alerts, SES for non-critical email
 - Limit increase delayed → Deploy across multiple accounts to stay within limits
 - Support case slow → Escalate through TAM or AWS account team
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How I'd Answer in an Interview

"Dependencies are the silent killer of enterprise projects. My approach is aggressive early action — I identify ALL cross-team dependencies in Week 1 and start the longest-lead items immediately, before we write a single line of Terraform. I maintain a RAID log reviewed weekly with stakeholders, and I escalate dependencies at risk BEFORE they become blockers, not after. The key is framing escalations in business impact terms — 'this delay costs us \$X per week' — not technical terms."

Question 5: Cutover / Go-Live Checklist

The EA Mindset

"A successful cutover is boring. It's boring because you've rehearsed it, tested it, and documented every step. Excitement during cutover means something went wrong."

Pre-Cutover Checklist (T-2 Weeks)

 Phase 1: Infrastructure Readiness

#	Check	Owner	Status
1	All target infrastructure provisioned and tested	Cloud Team	☐
2	Network connectivity verified (VPN/DC, all ports, all directions)	Network Team	☐
3	DNS entries prepared (not yet switched)	Network Team	☐
4	DNS TTL lowered to 60 seconds (48 hours before cutover)	Network Team	☐
5	Load balancers configured and health checks passing	Cloud Team	☐
6	Auto-scaling policies configured and tested	Cloud Team	☐
7	Backup and snapshot schedules configured	Cloud Team	☐
8	DR/failover tested (if applicable)	Cloud + SRE	☐

 Phase 2: Data Migration Readiness

#	Check	Owner	Status
9	Full data load completed	DBA Team	☐
10	CDC replication running and lag < 5 seconds	DBA Team	☐
11	Data validation passed (row counts, checksums, business rules)	DBA + App Team	☐
12	Database performance benchmarks match or exceed source	DBA Team	☐
13	Rollback procedure tested (can we revert to source DB?)	DBA Team	☐

 Phase 3: Application Readiness

#	Check	Owner	Status
14	Application deployed and running in cloud	Dev Team	☐

#	Check	Owner	Status
15	Configuration updated for cloud endpoints (DB, cache, queues)	Dev Team	☐
16	UAT sign-off received from business stakeholders	App Owner	☐
17	Performance/load test passed in cloud environment	QA Team	☐
18	Integration tests passed (all upstream/downstream connections)	Dev + QA	☐
19	SSL certificates installed and verified	Security	☐

Phase 4: Security Sign-Off

#	Check	Owner	Status
20	Security group rules reviewed and approved	Security Team	☐
21	IAM roles follow least-privilege principle	Security Team	☐
22	Encryption at rest enabled (all storage, databases)	Cloud Team	☐
23	Encryption in transit enabled (TLS 1.2+)	Cloud Team	☐
24	Vulnerability scan completed, no CRITICAL findings	Security Team	☐
25	Compliance sign-off received	GRC Team	☐
26	Penetration test completed (if required)	Security Team	☐

Phase 5: Monitoring & Observability

#	Check	Owner	Status
27	CloudWatch alarms configured for all critical metrics	SRE Team	☐
28	Dashboards created (infrastructure + application)	SRE Team	☐
29	Log aggregation working (application + access + audit logs)	SRE Team	☐

#	Check	Owner	Status
30	Alerting pipeline tested (alarm → SNS → PagerDuty/Slack)	SRE Team	☐
31	On-call schedule configured for post-cutover support	SRE Lead	☐

✅ Phase 6: Communication & Stakeholders

#	Check	Owner	Status
32	Cutover plan distributed to all stakeholders	Project Manager	☐
33	War room / bridge dial-in details shared	Project Manager	☐
34	Business communication sent (maintenance window notice)	Comms Team	☐
35	Rollback decision criteria documented and agreed	EA + App Owner	☐
36	Post-cutover support plan communicated	SRE Lead	☐

✅ Phase 7: Rollback Readiness

#	Check	Owner	Status
37	Rollback procedure documented step-by-step	Cloud Team	☐
38	Rollback tested in staging/DR environment	Cloud + DBA	☐
39	Rollback decision point defined (T+2 hours, T+4 hours)	EA	☐
40	On-prem environment kept running during hypercare period	Ops Team	☐

Go / No-Go Decision Matrix

Decision meeting: T-24 hours before cutover

Criterion	Go	No-Go
UAT sign-off	✓ Received	✗ Not received
Data validation	✓ All checks passed	✗ Discrepancies found
Performance test	✓ Meets or exceeds baseline	✗ Degradation > 20%
Security scan	✓ No CRITICAL findings	✗ Open CRITICAL findings
Network connectivity	✓ All paths tested	✗ Any path untested
Rollback plan	✓ Tested and documented	✗ Not tested
Key personnel available	✓ All on-call confirmed	✗ Key person unavailable
Business calendar	✓ No conflicting events	✗ Month-end, peak season

Decision authority: The EA recommends, but the **Application Owner** (business) makes the final Go/No-Go call. This is documented in the cutover minutes.

Cutover Night Execution Timeline (Example)

Friday 10:00 PM – Cutover begins

- └─ 10:00 PM: War room opens, roll call of all teams
- └─ 10:15 PM: Stop application writes to source database
- └─ 10:20 PM: Verify DMS CDC lag = 0 (all data replicated)
- └─ 10:30 PM: Run data validation scripts (row counts, checksums)
- └─ 10:45 PM: Switch application config to cloud database endpoint
- └─ 11:00 PM: Deploy application to cloud (or enable traffic to cloud)
- └─ 11:15 PM: Switch DNS to cloud load balancer
- └─ 11:30 PM: Smoke tests (critical user journeys)
- └─ 11:45 PM: Monitoring check (all dashboards green?)
- └─ DECISION POINT (midnight):
 - └─ All green → Proceed → Notify stakeholders "cutover successful"
 - └─ Issues found → Evaluate:
 - └─ Fixable in < 1 hour → Fix, retest
 - └─ Not fixable → ROLLBACK: switch DNS back, re-enable source DB

Saturday 12:00 AM – 8:00 AM – Hypercare monitoring

- └─ Engineer on-call monitoring all dashboards
- └─ Hourly status updates to war room
- └─ 8:00 AM: Morning standup – confirm stable

Monday 9:00 AM – Business validation

- └─ Business users confirm application working normally
- └─ Post-cutover sign-off

